DISCRETE MATH · THEORY OF GRAPHS AND TREES

Graph Explorer: Euler, Hamilton & Trees

Explore & Reflect

Name _____ Date _____ Period _____

Open **Graph Explorer: Euler, Hamilton & Trees** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A graph has degrees 3, 3, 2, 2, 2. How many edges does it have?

2. Can you draw the Königsberg bridges (degrees 5, 3, 3, 3) without lifting your pencil or repeating a bridge? Predict and explain.

2 Explore use the tool

1. Load Königsberg. Add one bridge so an Euler trail exists. Between which land masses must it go?

2. Load the Bowtie. Does it have an Euler circuit? A Hamiltonian cycle? Why?

3. Load C_5 and turn on Coloring. What is χ? Is there a triangle?

4. Load the weighted road network and run Kruskal. Record the edges taken, the edge skipped and the total.

3 Record use the tool and fill in the table

Graph facts

Load each graph and read the facts panel.

Graph	Vertices	Edges	Euler	Hamiltonian cycle?	χ
K_4					
K_5					
$K_{3,3}$					
Petersen					
House					

4 Explain what did you discover?

1. The sum of the degrees equals _____ times the number of edges.
2. A connected graph has an Euler circuit exactly when _____ degree is even, and an Euler trail (not circuit) when exactly _____ degrees are odd.
3. A connected graph with n vertices is a tree exactly when it has _____ edges.
4. Explain the difference between an Euler circuit and a Hamiltonian cycle, and why there is a simple test for one but not the other.

5 Apply a new situation, no tool

1. A tree has 12 vertices. How many edges? Can it have a vertex of degree 12?
2. Is there a simple graph with degrees 3, 3, 3, 1? Explain.
3. Exams A, B, C, D, E: pairs sharing a student are AB, AC, BC, BD, CD, DE. What is the fewest number of time slots?
4. K_6: how many edges? Euler circuit?